

Name:

Surname:

Matricola:

Module 1:

Exercise 1) [10 points]

The temperature T of a membrane of area $S = 2$ is a function of the external temperature T_e and of the heating exchanged with a pump $n(t)$ as defined by the following.

$$\frac{dT(t)}{dt} = 0.4(T_e(t) - T(t)) + n(t)/S$$

the exchanged heating is imposed by a tension ruled by a selector percentage $v(t)$

$$n(t) = 200 v(t)$$

We are interested in the temperature $T(t)$.

1. Determine the dynamic equations that define the system.
2. Classify the system.
3. Find an equilibrium point such that the valve opening is 10 %.
4. If the equilibrium point exists, determine, if possible, its stability.

Exercise 2) [11 points]

Given the following LTI System

$$\begin{cases} \dot{x}_1(t) = -x_1(t) - 3x_2(t) + u(t) \\ \dot{x}_2(t) = x_1(t) - x_2(t) - u(t) \\ y(t) = x_1(t) + x_2(t) \end{cases}$$

1. Define the transfer function.
2. Is the transfer function a solid approximation of the system? Motivate the answer.

Exercise 3) [11 points]

Consider the transfer function:

$$G(s) = \frac{100}{s^2 + 10.1s + 1}$$

1. determine the following indexes for the response to a 3 unit step
 - a. Equilibrium value
 - b. Settling time at 5%
2. Is it possible to have overshoot?
3. Evaluate the response to the following signal: $\cos(100t) + \cos(t/100)$

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Module 2:

Exercise 1) [8 points]

Consider the transfer function:

$$G(s) = \frac{100(s^2 + 1)}{s^2 - 4}$$

1. Draw the root locus
2. Find the gain such that the closed loop system has a pole in the origin

Exercise 2) [11 points for all requests, 8 points for 2 requests]

Given the following system

$$G(s) = \frac{2s}{(s + 100)(s + 1)}$$

Define a controller in a nominal scenario such that the controlled output step response addresses at least 2 of the following requests:

- Show no overshoot
- Has an error lower than 5%
- Has a setting time of at most 10 seconds

Exercise 3) [13 points for all requests, 9 points for 3 requests]

Given the following system

$$G(s) = \frac{100}{s(10s + 1)(s + 100)}$$

Define a controller in a real scenario such that the controlled output step response addresses at least 3 of the following requests:

- Show an overshoot lower than 15%;
- Has an error lower than 5%;
- Has a setting time of at most 10 seconds;
- The measurement noise has an attenuation of at least 20 db.

Knowing that

- The measurement noise shows significant components only for frequency greater than 10KHz
- There is no disturbance

MOD 2

EXERCISE 1

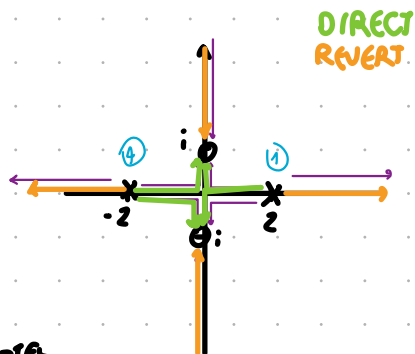
$$G(s) = \frac{100(s^2 + 1)}{s^2 - 4}$$

DRAW ROOT LOCUS

• POLES = $\{-2; +2\}$

• ZEROS = $\{-i; +i\}$

$\sigma = \frac{-2 - 2}{2} = -2 \rightarrow$ NO ASYMPTOTE



$$s^2 + 1 = 0 \rightarrow s^2 = -1$$

$$s = \pm i$$

DISTANCES ARE POSITIVE

$$P = \frac{\prod m_i}{\prod \lambda_i} = \frac{(2)(+2)}{(+i)(-i)} = \frac{+4}{+1} = +4$$

← WE EXPECT A POSITIVE GAIN SINCE DL

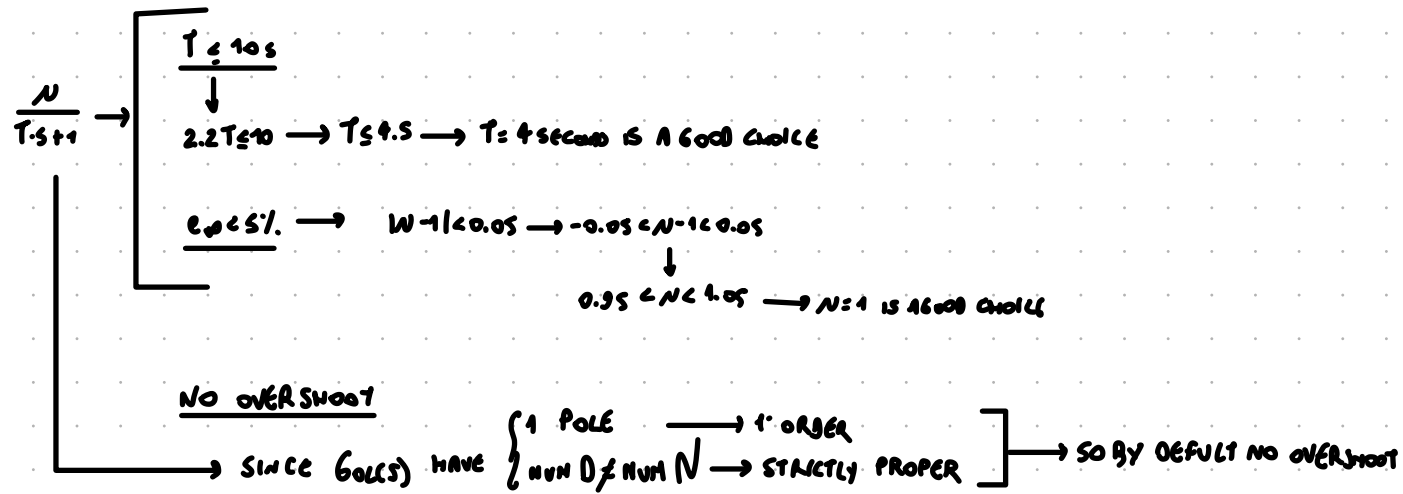
EXERCISE 2 (RIFATTO)

$$G(s) = \frac{2s}{(s+100)(s+1)} \longrightarrow G^{-1} \text{ STABLE} \longrightarrow \begin{cases} \bullet \text{ I USE POLE PLACEMENT (OPEN LOOP)} \\ \bullet \text{ THE CONTROLLED WOULD BE } C = \frac{G_{OL}(s)}{G(s)} \text{ WHERE } G_{OL}(s) = \frac{N}{T \cdot s + 1} \end{cases}$$

• NO OVERTHOOT

• ERROR LOWER 5% $\longrightarrow e_{ss} < 5\%$

• SETTLING TIME AT MOST 10 SECONDS $\longrightarrow T_s \leq 10 \text{ s}$



$$G_{OL}(s) = \frac{N}{T \cdot s + 1} = \frac{1}{4 \cdot s + 1}$$

$$C = \frac{G_{OL}}{G(s)} = \frac{1}{4 \cdot s + 1} \cdot \frac{(s+100)(s+1)}{2s}$$

• FEASIBLE $\checkmark \longrightarrow \deg D \geq \deg N$ (3) 1
 • STABLE $\checkmark \longrightarrow \forall p \leq 0$ $\longrightarrow$ IF NOT FEASIBLE ADD A POLE

$$\frac{1}{4 \cdot s + 1} \cdot \frac{2s(s+1)^2}{(s+100)(s+1)} \longrightarrow \frac{1}{(5/10^3 + 1)}$$

EXERCISE 3 RIFATTO

$$G(s) = \frac{100}{s(10s+1)(5s+100)}$$

- OVERSHOOT LOWER THAN 15% $\rightarrow \zeta < 15\%$
- ERROR LOWER 5% $\rightarrow e_p < 5\%$
- SETTLING TIME AT MOST 10 S $\rightarrow T \leq 10 \text{ seconds}$
- NOISE ATTENUATION AT LEAST 20 dB $\rightarrow ?$

$$e_p < 5\%$$

$\rightarrow$ since $q=1$ ^{zeros in P - zero in Z = 1-0} NO ERROR!!

$$T < 10 \text{ s}$$

$\rightarrow 2.2T < 10 \rightarrow T = 4.5 \text{ sec.} \rightarrow T = 4 \text{ sec.}$ IS A GOOD CHOICE